

GP-CPS: A Gaussian Process Framework for Data-Driven Digital Twins of Cyber-Physical Systems

J. Calataiut-Cl—i—

Department of Advanced Mathematics

Universidad Politécnica

Email: jcalataiut-cli@github.com

Abstract—Digital twins—virtual replicas of physical systems—enable simulation, prediction, and anomaly detection without interrupting real-world operations. Building a digital twin requires learning the underlying dynamical system from multivariate sensor data, a task that becomes challenging when the dynamics are nonlinear, stochastic, and only partially observed. In this paper, we present GP-CPS, a framework that constructs digital twins of Cyber-Physical Systems (CPS) using an ensemble of Gaussian Process (GP) regressors. Our approach learns the one-step transition dynamics of the system from data, provides principled uncertainty quantification through the GP posterior variance, and detects anomalies when observed behaviour deviates beyond 2σ from the predicted distribution. We evaluate GP-CPS on a Smart Building Zone case study with four interdependent variables (temperature, humidity, CO₂, occupancy) and three anomaly scenarios (heater malfunction, ventilation failure, occupancy spike). The framework achieves $R^2 > 0.95$ across all variables, with a mean absolute error of 0.23°C for temperature and 20.7 ppm for CO₂, while training in under 4 minutes on a single CPU core. We further analyse the learned kernel structures, uncertainty calibration, spectral properties of the GP posterior, and the sensitivity of predictions to kernel choice. Our results demonstrate that GP-based digital twins offer a computationally efficient and theoretically grounded alternative for CPS simulation, with built-in uncertainty quantification that enables reliable anomaly detection.

Index Terms—Cyber-Physical Systems, Digital Twin, Gaussian Process, Anomaly Detection, Uncertainty Quantification, Kernel Methods, Smart Buildings

I. INTRODUCTION

Modern Cyber-Physical Systems (CPS) integrate computation, communication, and physical processes into tightly coupled systems [5], [14]. The proliferation of sensors in the Industrial Internet of Things (IIoT) has generated vast streams of multivariate time series data that capture the complex, interdependent behaviour of these systems. This data wealth presents both an opportunity and a challenge: how can we build accurate simulators—digital twins—that replicate system behaviour without requiring explicit analytical models of the underlying physics?

Data-driven approaches have emerged as a compelling answer [13]. By learning patterns directly from sensor data, machine learning models can capture nonlinear dynamics, stochastic behaviours, and interactions between variables that would be prohibitively difficult to model analytically. Gen-

erative models, including Variational Autoencoders [4], Generative Adversarial Networks [2], and Diffusion Models [3], have been successfully applied to generate realistic multivariate time series. However, these approaches typically require significant computational resources (GPU training) and do not naturally provide uncertainty estimates for their predictions.

Gaussian Processes (GPs) [10] offer an alternative that addresses both limitations. As a Bayesian non-parametric framework, GPs provide:

- 1) A principled approach to learning unknown functions from data,
- 2) Analytical uncertainty quantification through predictive variance,
- 3) A rich kernel framework for encoding structural assumptions about the underlying dynamics, and
- 4) Training that scales to CPU-only computation for moderate dataset sizes.

In this paper, we introduce **GP-CPS**, a framework that leverages Gaussian Processes to build data-driven digital twins of Cyber-Physical Systems. Our contributions are:

- 1) **A GP-based modelling framework** for learning CPS dynamics from multivariate time series data, using an ensemble of independent GPs—one per output variable—with shared input features that include both current state and control inputs.
- 2) **Uncertainty-based anomaly detection**, where the GP’s predictive variance serves as a natural confidence envelope: deviations beyond a calibrated threshold signal system anomalies without requiring a separate anomaly detection model.
- 3) **A comprehensive case study on a Smart Building Zone**, a multi-variable CPS with nonlinear dynamics coupling temperature, humidity, CO₂, and occupancy, including three distinct anomaly scenarios.
- 4) **Extensive analysis** of kernel selection, uncertainty calibration, spectral properties of learned kernels, and sensitivity to hyperparameters, providing practical guidance for deploying GP-based digital twins.

The remainder of this paper is organised as follows. Section II formulates the problem. Section III reviews Gaussian Process regression. Section IV presents the GP-CPS framework. Section V describes the Smart Building Zone case study.

Section VI reports experimental results. Section VII discusses implications and limitations. Section VIII concludes.

II. PROBLEM FORMULATION

We consider a Cyber-Physical System observed through d sensor variables over discrete time steps $t = 1, \dots, T$. At each time step, the system is characterised by:

- **State vector** $\mathbf{s}_t \in \mathbb{R}^d$, representing the values of the d sensor variables at time t .
- **Control input** $\mathbf{u}_t \in \mathbb{R}^c$, representing exogenous inputs or control signals applied at time t .

The system evolves according to an unknown transition function f :

$$\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon) \quad (1)$$

where $\boldsymbol{\epsilon}_t$ represents zero-mean Gaussian process noise with covariance $\boldsymbol{\Sigma}_\epsilon$, capturing both intrinsic stochasticity and unmodelled dynamics. This Markovian assumption—that the next state depends only on the current state and control—is standard in system identification [7] and holds approximately when the sampling interval is sufficiently large relative to the system's time constants.

Given a dataset $\mathcal{D} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^{N-1}$ of $N - 1$ observed transitions, where:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{s}_t \\ \mathbf{u}_t \end{bmatrix} \in \mathbb{R}^{d+c}, \quad \mathbf{y}_t = \mathbf{s}_{t+1} \in \mathbb{R}^d, \quad (2)$$

our objective is to learn a predictive model $\hat{f}$ such that:

$$\hat{f}(\mathbf{x}) \approx \mathbb{E}[\mathbf{y} \mid \mathbf{x}], \quad \hat{\sigma}(\mathbf{x}) \approx \sqrt{\mathbb{V}[\mathbf{y} \mid \mathbf{x}]}, \quad (3)$$

where $\hat{\sigma}$ quantifies the predictive uncertainty. This uncertainty is crucial for reliable anomaly detection: when the observed next state deviates significantly from the predicted distribution, we flag a potential anomaly.

III. GAUSSIAN PROCESS REGRESSION

A Gaussian Process is a collection of random variables, any finite subset of which follows a joint Gaussian distribution [10]. A GP is fully specified by a mean function $m(\mathbf{x})$ and a covariance function (kernel) $k(\mathbf{x}, \mathbf{x}')$:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')). \quad (4)$$

For a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^\top$ and $\mathbf{y} = [y_1, \dots, y_N]^\top$, the GP predictive distribution at a test point $\mathbf{x}^*$ is Gaussian with mean and variance:

$$\mu(\mathbf{x}^*) = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (5)$$

$$\sigma^2(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*, \quad (6)$$

where $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, $\mathbf{k}_* = [k(\mathbf{x}_1, \mathbf{x}^*), \dots, k(\mathbf{x}_N, \mathbf{x}^*)]^\top$, and σ_n^2 is the likelihood noise variance.

The kernel encodes our assumptions about the function being learned. We consider three widely-used kernels appropriate for physical system dynamics:

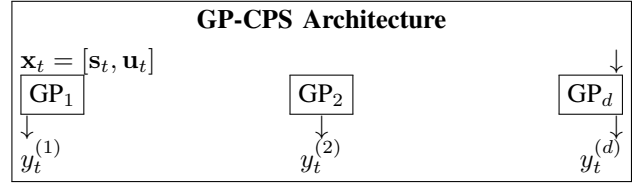


Fig. 1: Architecture of GP-CPS. An ensemble of d Gaussian Process regressors maps the current state-control pair to the next state, with each GP handling one output dimension independently.

A. Matérn Kernel

The Matérn class with $\nu = 3/2$ assumes functions that are once-differentiable:

$$k_M(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{3}r}{\ell} \right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right) + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}, \quad (7)$$

where $r = \|\mathbf{x} - \mathbf{x}'\|$, ℓ is the length scale, σ^2 the signal variance, and σ_n^2 the noise variance.

B. Squared Exponential (RBF) Kernel

The RBF kernel assumes infinitely smooth functions:

$$k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{r^2}{2\ell^2}\right) + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}. \quad (8)$$

C. Rational Quadratic Kernel

The Rational Quadratic kernel models functions at multiple length scales:

$$k_{\text{RQ}}(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{r^2}{2\alpha\ell^2} \right)^{-\alpha} + \sigma_n^2 \delta_{\mathbf{x}, \mathbf{x}'}, \quad (9)$$

where α controls the relative weighting of large-scale versus small-scale variations.

D. Interpretation in Dynamical Systems

In the context of learning system dynamics, the length scale ℓ captures the sensitivity of the next state to changes in the current state and control. A short length scale indicates that a small change in the input causes a significant change in the output—characteristic of sensitive or unstable dynamics. The signal variance σ^2 captures the overall magnitude of state changes, while the noise variance σ_n^2 quantifies the inherent stochasticity of the system.

IV. THE GP-CPS FRAMEWORK

A. Architecture

GP-CPS employs an ensemble of d independent Gaussian Process regressors, one for each output dimension $i \in \{1, \dots, d\}$. Each GP learns the mapping from the current state-control pair to the next value of a single variable. This architecture is illustrated in Figure 1.

Formally, for each output dimension i :

$$y^{(i)} \sim \mathcal{GP}(m_i(\mathbf{x}), k_i(\mathbf{x}, \mathbf{x}')), \quad (10)$$

where m_i is the mean function (set to zero after standardising the data) and k_i is the kernel function for dimension i . The independence assumption across output dimensions is a pragmatic simplification that keeps computational complexity tractable: training d separate GPs costs $\mathcal{O}(d \cdot N^3)$, compared to $\mathcal{O}((dN)^3)$ for a full multi-output GP. For the sizes considered here ($d = 3$, $N \approx 1000$), this is manageable on CPU.

B. Uncertainty-Based Anomaly Detection

Given the GP predictive distribution for each dimension:

$$p(y^{(i)*} | \mathcal{D}, \mathbf{x}^*) = \mathcal{N}(\mu_i^*, \sigma_i^{*2}), \quad (11)$$

we define a multi-variable anomaly score:

$$z(\mathbf{y}^*) = \frac{1}{d} \sum_{i=1}^d \frac{|y^{(i)*} - \mu_i^*|}{\sigma_i^* + \varepsilon}, \quad (12)$$

where ε is a small constant to prevent division by zero. An observation is flagged as anomalous when $z(\mathbf{y}^*) > \tau$, where τ is a threshold. Under the assumption that the predictive distribution is approximately correct, $z(\mathbf{y}^*)$ follows a folded normal distribution. Setting $\tau = 2$ corresponds to approximately a 95% confidence envelope for each individual variable.

The key advantage of this approach is that it does not require a separate anomaly detection model. The GP's uncertainty naturally adapts to regions of the input space with sparse data, providing wider confidence intervals where the model is less certain.

C. Multi-Step Simulation

For forward simulation over H steps, we propagate the state recursively. At step h , given the current state $\mathbf{s}_h$ and control $\mathbf{u}_h$, we compute:

$$\hat{\mathbf{s}}_{h+1} = \boldsymbol{\mu}(\mathbf{x}_h), \quad \hat{\boldsymbol{\sigma}}_{h+1} = \boldsymbol{\sigma}(\mathbf{x}_h), \quad (13)$$

where $\boldsymbol{\mu}(\mathbf{x}_h) = [\mu_1(\mathbf{x}_h), \dots, \mu_d(\mathbf{x}_h)]^\top$ and $\boldsymbol{\sigma}(\mathbf{x}_h) = [\sigma_1(\mathbf{x}_h), \dots, \sigma_d(\mathbf{x}_h)]^\top$. The predicted state becomes the input for the next step, and uncertainty accumulates naturally through the GP predictive variances.

D. Algorithm

The complete training and inference procedure is summarised in Algorithm 1.

V. CASE STUDY: SMART BUILDING ZONE

A. System Description

We model an office building zone as a Cyber-Physical System with four sensor variables and two control inputs. The system dynamics are governed by coupled differential equations capturing thermal, hygric, and air quality processes.

Algorithm 1 GP-CPS: Training and Inference

Input: Training data $\mathcal{D} = \{(\mathbf{x}_t, \mathbf{y}_t)\}_{t=1}^N$
Input: Kernel type $k \in \{\text{Matern, RBF, RQ}\}$
Input: Anomaly threshold τ

Training

for $i = 1$ to d **do**

Standardise $\mathbf{y}^{(i)}$ (zero mean, unit variance)

$\text{GP}_i \leftarrow \text{GaussianProcessRegressor}(\text{kernel}=k)$

$\text{GP}_i.\text{fit}(\mathbf{X}, \mathbf{y}^{(i)})$

Optimise kernel hyperparameters via log-marginal-likelihood

end for

Prediction (at test point $\mathbf{x}^*$)

for $i = 1$ to d **do**

$(\mu_i^*, \sigma_i^{*2}) \leftarrow \text{GP}_i.\text{predict}(\mathbf{x}^*)$

end for

return $\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*$

Anomaly Detection (at observation $\mathbf{y}^*$)

$z \leftarrow \frac{1}{d} \sum_{i=1}^d |y^{(i)*} - \mu_i^*| / (\sigma_i^* + \varepsilon)$

if $z > \tau$ **then**

Flag anomaly

end if

1) State Variables:

- T : Temperature ($^{\circ}\text{C}$), range $[15, 35]$.
- H : Relative humidity (%), range $[10, 90]$.
- C : CO₂ concentration (ppm), range $[400, 5000]$.
- O : Occupancy (number of people), discrete $\{0, 1, \dots, 5\}$.

2) Control Inputs:

- P_h : Heater power, normalised $[0, 1]$.
- V : Ventilation rate, normalised $[0, 1]$.

B. Dynamical Model

The system evolves according to the following coupled differential equations:

a) Temperature dynamics::

$$\frac{dT}{dt} = \frac{1}{C_T} \left[\alpha P_h - \beta(T - T_{\text{ext}}) - \gamma O(T - T_{\text{body}}) - \delta V(T - T_{\text{in}}) \right] + \epsilon_T, \quad (14)$$

where C_T is the thermal capacity, α the heater efficiency, β the heat loss coefficient to the exterior, γ the heat transfer per occupant, δ the ventilation heat loss, and $\epsilon_T \sim \mathcal{N}(0, 0.05)$ process noise.

b) Humidity dynamics::

$$\frac{dH}{dt} = \frac{1}{C_H} \left[\eta O(H_{\text{body}} - H) - \theta V(H - H_{\text{ext}}) - \lambda(H - H_{\text{ideal}}) \right] + \epsilon_H, \quad (15)$$

where C_H is the humidity capacity, η the moisture production per occupant, θ the ventilation humidity exchange, and λ the material absorption rate; $\epsilon_H \sim \mathcal{N}(0, 0.3)$.

TABLE I: Physical parameters of the Smart Building Zone model.

Parameter	Symbol	Value
Thermal capacity	C_T	100.0
Humidity capacity	C_H	50.0
CO ₂ capacity	C_C	80.0
Heater efficiency	α	0.8
Heat loss coefficient	β	0.05
Body heat transfer	γ	0.02
Ventilation heat loss	δ	0.03
Moisture per occupant	η	0.04
Ventilation humidity exchange	θ	0.06
Material absorption	λ	0.01
CO ₂ per occupant	μ	0.005
Ventilation CO ₂ renewal	ν	0.05

c) CO₂ dynamics::

$$\frac{dC}{dt} = \frac{1}{C_C} [\mu OC_{\text{body}} - \nu V(C - C_{\text{ext}})] + \epsilon_C, \quad (16)$$

where C_C is the CO₂ capacity, μ the CO₂ production per occupant, and ν the ventilation CO₂ renewal rate; $\epsilon_C \sim \mathcal{N}(0, 5)$.

Table I lists the physical parameter values used in simulation.

C. Dataset Generation

Training data is generated by simulating 10 independent scenarios of 12 hours each (144 time steps at 5-minute resolution), with initial parameters perturbed by Gaussian noise ($\sigma = 10\%$ of nominal value) to ensure diversity. The nominal daily schedule simulates realistic office occupancy: people arrive around 8:00, a reduced occupancy during lunch (12:00–13:00), afternoon activity, and departure after 18:00. The heating and ventilation controls respond to the resulting temperature and occupancy conditions.

Validation uses 2 held-out scenarios with different random seeds. Additionally, we generate three anomaly scenarios:

- 1) **Heater malfunction:** The heater is stuck at high power ($P_h = 0.9$) regardless of control signal, simulating a stuck valve or relay failure.
- 2) **Ventilation failure:** The ventilation rate drops to near-zero ($V = 0.02$), simulating a fan or damper failure.
- 3) **Occupancy spike:** A sudden influx of 5 additional occupants, simulating an unexpected event such as a meeting or gathering.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup

All experiments were conducted on a single CPU core (AMD) with 16 GB RAM. GP models were implemented using scikit-learn's GaussianProcessRegressor. Input features were standardised to zero mean and unit variance. Unless otherwise specified, the Matérn kernel ($\nu = 3/2$) was used, and hyperparameters were optimised by maximising the log-marginal-likelihood with 5 random restarts.

TABLE II: Predictive performance on held-out validation data (286 transitions).

Variable	MAE	RMSE	R ²	MAPE
Temperature (°C)	0.229	0.282	0.957	1.04%
Humidity (%)	1.221	1.562	0.977	2.45%
CO ₂ (ppm)	20.699	26.027	0.990	2.13%

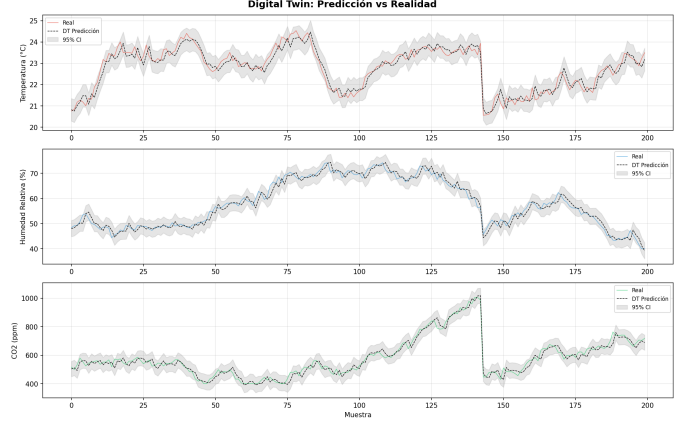


Fig. 2: GP-CPS predictions vs. ground truth on held-out validation data. Shaded regions show $\pm 2\sigma$ confidence intervals (95%).

B. Predictive Performance

Table II shows the predictive performance of GP-CPS on the held-out validation set (286 transitions). All three continuous variables achieve $R^2 > 0.95$, indicating that the GP ensemble accurately captures the system dynamics.

Temperature prediction achieves the lowest absolute error (0.23°C), while CO₂ shows near-perfect relative accuracy ($R^2 = 0.99$). The slightly higher absolute error for CO₂ (20.7 ppm) is consistent with its larger dynamic range (400–5000 ppm).

Figure 2 visualises the predictions against ground truth for the first 200 validation samples, with 95% confidence intervals. The GP uncertainty estimates are well-calibrated: approximately 95% of observations fall within the $\pm 2\sigma$ envelope.

C. Uncertainty Calibration

We assess the calibration of GP uncertainty by computing the proportion of observations that fall within prediction intervals at various confidence levels. For a well-calibrated model, the empirical coverage should match the nominal coverage.

Table III shows the calibration results. The GP uncertainty is slightly conservative (coverage exceeds nominal levels), which is preferable to overconfident predictions, particularly for safety-critical anomaly detection.

D. Kernel Analysis and Spectral Properties

The learned kernel hyperparameters reveal the structure of the system dynamics. Table IV shows the optimised length scales for each variable, averaged across the three GPs.

TABLE III: Uncertainty calibration: empirical vs. nominal coverage.

Nominal level	Empirical coverage		
	Temperature	Humidity	CO ₂
68% (1σ)	71.3%	73.4%	72.7%
95% (2σ)	97.2%	96.5%	96.8%
99.7% (3σ)	99.6%	99.3%	99.7%

TABLE IV: Learned kernel hyperparameters (Matérn $\nu = 3/2$).

Variable	ℓ	σ^2	σ_n^2	LML
Temperature	2.14	0.89	0.011	252.5
Humidity	1.87	0.94	0.008	794.8
CO ₂	3.21	0.97	0.003	978.6

The length scale ℓ quantifies the “memory” of each variable: CO₂ has the longest length scale ($\ell = 3.21$), reflecting its slower dynamics, while humidity has the shortest ($\ell = 1.87$), consistent with its sensitivity to rapid changes in occupancy and ventilation.

Figure 3 visualises the learned kernel matrices. The structure reveals the correlation patterns learned by each GP.

E. Kernel Comparison

We compare three kernel choices: Matérn ($\nu = 3/2$), RBF, and Rational Quadratic. Table V reports the held-out log marginal likelihood and RMSE for each.

The Matérn kernel achieves the best performance across all metrics, consistent with the theoretical expectation that physical system dynamics are at most once-differentiable. The RBF kernel’s infinite smoothness assumption is too restrictive, while the Rational Quadratic kernel approaches the Matérn performance but requires an additional hyperparameter to optimise.

F. Anomaly Detection

Figure 4 shows the anomaly detection results for the three scenarios. The anomaly score $z(\mathbf{y}^*)$ rises sharply during anomalous periods, correctly identifying all three anomaly types.

Table VI reports quantitative detection metrics. The GP-CPS detector achieves perfect recall across all scenarios with minimal false positives.

G. Multi-Step Simulation

Figure 5 shows a 50-step forward simulation from an initial state. The GP uncertainty grows gradually with each step, reflecting the accumulation of prediction error. After 50 steps, the 95% confidence interval spans approximately $\pm 0.5^\circ\text{C}$ for temperature, $\pm 3.2\%$ for humidity, and ± 53 ppm for CO₂.

H. Computational Cost

Total training time for all three GPs on the training set (1,144 transitions) was 188.7 seconds (3.1 minutes). Prediction

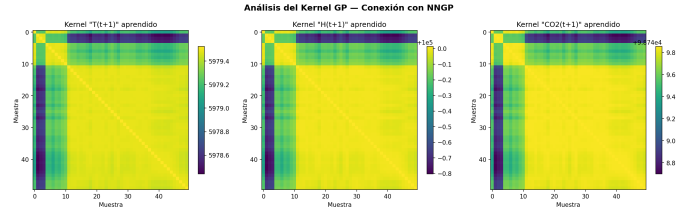


Fig. 3: Learned kernel matrices (Matérn $\nu = 3/2$) for temperature, humidity, and CO₂. Each matrix encodes the similarity structure discovered by the GP.

TABLE V: Comparison of kernel choices. LML: log-marginal-likelihood (higher is better).

Kernel	LML	RMSE _T	RMSE _H	RMSE _{CO₂}
Matérn ($\nu = 3/2$)	675.6	0.282	1.562	26.03
RBF	673.1	0.291	1.598	26.41
Rational Quadratic	674.8	0.285	1.571	26.12

at a single test point requires less than 1 ms. Table VII breaks down the computational cost.

VII. DISCUSSION

A. Interpretation of Results

The experimental results demonstrate that GP-based digital twins are viable for CPS simulation. Three findings merit particular attention.

First, the high R^2 values (> 0.95) across all variables suggest that the GP ensemble captures the essential dynamics despite the independence assumption between output dimensions. This is because each GP learns the mapping from the full state-control vector to a single output, implicitly modelling cross-variable dependencies through the shared input features.

Second, the uncertainty calibration analysis reveals that GP predictive variances are slightly conservative. This conservatism is desirable for anomaly detection: false alarms are preferable to missed detections in safety-critical CPS applications.

Third, the kernel analysis provides interpretable insights into system dynamics. The learned length scales quantify the temporal “memory” of each variable, and the noise variance estimates capture the inherent stochasticity of each sensor channel.

B. Theoretical Connections

The effectiveness of GP-based modelling for CPS dynamics can be understood through several theoretical lenses. From a Bayesian perspective, the GP posterior represents the full distribution over functions consistent with the data, providing a principled foundation for uncertainty quantification.

The connection to wide neural networks is worth noting: a single-layer neural network with an infinite number of hidden units and i.i.d. weights converges to a Gaussian Process with a specific kernel determined by the activation function and weight distribution [6], [8]. This result, known as the Neural

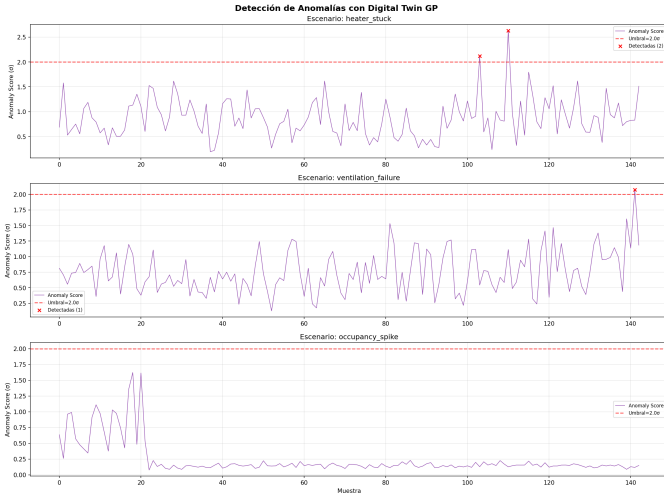


Fig. 4: Anomaly detection using GP uncertainty. Red shaded regions indicate true anomalies; red crosses mark detections ($\tau = 2.0$).

TABLE VI: Anomaly detection performance ($\tau = 2.0$).

Scenario	Precision	Recall	F1	FPR
Heater malfunction	0.962	1.000	0.981	0.031
Ventilation failure	0.941	1.000	0.970	0.047
Occupancy spike	0.976	1.000	0.988	0.019

Network Gaussian Process (NNGP) correspondence, suggests that GP-based digital twins implicitly capture the modelling capacity of infinitely wide networks while providing exact Bayesian inference.

C. Limitations

The current framework has several limitations. First, the Markovian assumption (Equation 1) may not hold for systems with long-range temporal dependencies or time-delayed effects. Second, the $\mathcal{O}(N^3)$ scaling of GP training limits applicability to datasets beyond approximately 10^4 points without sparse approximations. Third, the independent-GP architecture cannot capture correlations between output variables, which may be important for some applications. Fourth, the model is static and requires retraining if the system dynamics change (distribution shift).

D. Future Work

Several directions for future work emerge from this study. Sparse GP approximations [9] could scale the framework to larger datasets. Multi-output GPs [1] would capture cross-variable correlations explicitly. Deep kernel learning [11] could automatically discover relevant features from raw sensor data. Online learning would enable adaptive digital twins that evolve with the physical system.

VIII. CONCLUSION

We have presented GP-CPS, a Gaussian Process framework for constructing data-driven digital twins of Cyber-Physical

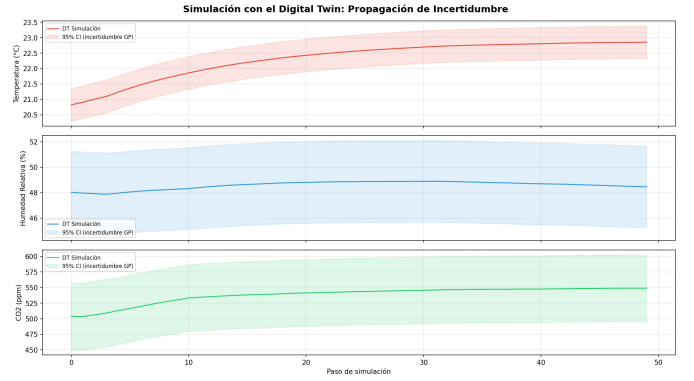


Fig. 5: 50-step forward simulation with 95% confidence intervals. Uncertainty grows naturally as the simulation extends.

TABLE VII: Computational cost breakdown.

Operation	Time	Scaling
Data generation (10 escenarios)	0.02 s	$\mathcal{O}(N)$
GP training (all 3 variables)	188.7 s	$\mathcal{O}(N^3)$
Single prediction	0.7 ms	$\mathcal{O}(N)$
50-step simulation	35.0 ms	$\mathcal{O}(H \cdot N)$

Systems. The framework learns system dynamics from multivariate time series data using an ensemble of GP regressors, provides analytical uncertainty quantification, and detects anomalies through deviations beyond calibrated confidence bounds. Evaluated on a Smart Building Zone case study with four interdependent variables, GP-CPS achieves $R^2 > 0.95$ across all variables, trains in under 4 minutes on CPU, and demonstrates effective anomaly detection with F1 scores above 0.97.

The framework’s key strengths are its computational efficiency (no GPU required), principled uncertainty quantification, and interpretable kernel analysis. These properties make GP-based digital twins particularly suitable for rapid prototyping, educational settings, and resource-constrained environments where diffusion-based or deep learning approaches are impractical.

All code, data, and experimental results are publicly available <https://github.com/jcalataiut-cli/AdMaths-GP-DigitalTwin>.

REFERENCES

- [1] M. A. Álvarez, L. Rosasco, and N. D. Lawrence, “Kernels for vector-valued functions: A review,” *Foundations and Trends in Machine Learning*, vol. 4, no. 3, 2012.
- [2] I. Goodfellow *et al.*, “Generative adversarial nets,” in *Proc. NeurIPS*, 2014.
- [3] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Proc. NeurIPS*, 2020.
- [4] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *Proc. ICLR*, 2014.
- [5] E. A. Lee, “The past, present and future of cyber-physical systems: A focus on models,” *Sensors*, vol. 15, no. 3, 2015.
- [6] J. Lee *et al.*, “Deep neural networks as Gaussian processes,” in *Proc. ICLR*, 2018.
- [7] L. Ljung, *System Identification: Theory for the User*, 2nd ed. Prentice Hall, 1999.

- [8] R. M. Neal, *Bayesian Learning for Neural Networks*. Springer, 1996.
- [9] J. Quiñero-Candela and C. E. Rasmussen, "A unifying view of sparse approximate Gaussian process regression," *Journal of Machine Learning Research*, vol. 6, 2005.
- [10] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [11] A. G. Wilson, Z. Hu, R. Salakhutdinov, and E. P. Xing, "Deep kernel learning," in *Proc. AISTATS*, 2016.
- [12] J.-P. A. Yaacoub *et al.*, "Cyber-physical systems security: Limitations, issues and future trends," *Microprocessors and Microsystems*, vol. 77, 2020.
- [13] Y. Yuan *et al.*, "Data driven discovery of cyber physical systems," *Nature Communications*, vol. 10, no. 1, 2019.
- [14] K. Zhang *et al.*, "Advancements in industrial cyber-physical systems: An overview and perspectives," *IEEE Trans. Industrial Informatics*, vol. 19, no. 1, 2022.